# Containerization-and-DevOps

# 🚀 Class Task

## 🛠️ Tools Used

- Docker
- k3d
- kubectl
- Kubernetes

## ⚙️ Steps

### Check k3d Commands

```
k3d
k3d cluster
```

### List Existing Clusters

```
k3d cluster list
```

### Start Cluster

```
k3d cluster start mycluster
```

### Verify Cluster Again

```
k3d cluster list
```

## Check Pods (Initial State)

```
kubectl get pods
```

## Create Pod

```
kubectl run nginx --image=nginx
```

## Describe Pod

```
kubectl describe pod nginx
```

## Check Pod Logs

```
kubectl logs nginx
```

## Create Deployment

```
kubectl create deployment web --image=nginx
```

## Scale Deployment

```
kubectl scale deployment web --replicas=3
```

## Expose Deployment

```
kubectl expose deployment web --port=80 --type=NodePort
```

### Check Services

```
kubectl get services
```

### Check Pods

```
kubectl get pods
```

### Delete Pod

```
kubectl delete pod nginx
```

## ⚠️ Notes

- Initially no pods were present in default namespace
- A single nginx pod was created and verified
- Deployment created multiple replicas (3 pods)
- NodePort service exposes the application externally
- Typo in command ( `sclae` ) corrected to `scale` 😅

## ✅ Outcome

- Kubernetes cluster successfully managed using k3d
- Pod creation, logs, and inspection verified
- Deployment scaled to multiple replicas
- Service exposed using NodePort
- Pod deletion tested successfully

## 📸 Screenshots

> Add your screenshots here:

```
sristee@Sristee:/mnt/c/Users/srist$ k3d
Usage:
  k3d [flags]
  k3d [command]

Available Commands:
  cluster      Manage cluster(s)
  completion   Generate completion scripts for [bash, zsh, fish, powershell | psh]
  config       Work with config file(s)
  help         Help about any command
  image        Handle container images.
  kubeconfig   Manage kubeconfig(s)
  node         Manage node(s)
  registry     Manage registry/registries
  version      Show k3d and default k3s version

Flags:
  -h, --help         help for k3d
      --timestamps   Enable Log timestamps
      --trace        Enable super verbose output (trace logging)
      --verbose      Enable verbose output (debug logging)
      --version      Show k3d and default k3s version

Use "k3d [command] --help" for more information about a command.
sristee@Sristee:/mnt/c/Users/srist$ k3d cluster
Manage cluster(s)

Usage:
  k3d cluster [flags]
  k3d cluster [command]

Available Commands:
  create      Create a new cluster
```

```
sristee@Sristee:/mnt/c/Users/srist$ k3d cluster list
NAME        SERVERS   AGENTS   LOADBALANCER
mycluster   1/1       0/0      true
sristee@Sristee:/mnt/c/Users/srist$ k3d cluster start mycluster
INFO[0000] Using the k3d-tools node to gather environment information
INFO[0000] Starting new tools node...
INFO[0000] Starting node 'k3d-mycluster-tools'
INFO[0000] HostIP: using network gateway 172.20.0.1 address
INFO[0000] Starting cluster 'mycluster'
INFO[0000] All servers already running.
INFO[0000] All agents already running.
INFO[0000] All helpers already running.
INFO[0000] Started cluster 'mycluster'
sristee@Sristee:/mnt/c/Users/srist$ k3d cluster list
NAME        SERVERS   AGENTS   LOADBALANCER
mycluster   1/1       0/0      true
sristee@Sristee:/mnt/c/Users/srist$ kubectl get pod
No resources found in default namespace.
sristee@Sristee:/mnt/c/Users/srist$ kubectl get pods
No resources found in default namespace.
sristee@Sristee:/mnt/c/Users/srist$ kubectl run nginx --image=nginx
pod/nginx created
sristee@Sristee:/mnt/c/Users/srist$ kubectl describe pod nginx
Name:             nginx
Namespace:        default
Priority:         0
Service Account:  default
Node:             k3d-mycluster-server-0/172.20.0.2
Start Time:       Thu, 19 Mar 2026 04:53:54 +0000
Labels:           run=nginx
Annotations:      <none>
Status:           Running
IP:               10.42.0.9
```

```
  Normal  Created    4m4s   kubelet            spec.containers{nginx}: Created container nginx
  Normal  Started    4m4s   kubelet            spec.containers{nginx}: Started container nginx
sristee@Sristee:/mnt/c/Users/srist$ kubectl logs nginx
/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
/docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
/docker-entrypoint.sh: Configuration complete; ready for start up
2026/03/19 04:54:26 [notice] 1#1: using the "epoll" event method
2026/03/19 04:54:26 [notice] 1#1: nginx/1.29.6
2026/03/19 04:54:26 [notice] 1#1: built by gcc 14.2.0 (Debian 14.2.0-19)
2026/03/19 04:54:26 [notice] 1#1: OS: Linux 6.6.87.2-microsoft-standard-WSL2
2026/03/19 04:54:26 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1048576:1048576
2026/03/19 04:54:26 [notice] 1#1: start worker processes
2026/03/19 04:54:26 [notice] 1#1: start worker process 29
2026/03/19 04:54:26 [notice] 1#1: start worker process 30
2026/03/19 04:54:26 [notice] 1#1: start worker process 31
2026/03/19 04:54:26 [notice] 1#1: start worker process 32
2026/03/19 04:54:26 [notice] 1#1: start worker process 33
2026/03/19 04:54:26 [notice] 1#1: start worker process 34
2026/03/19 04:54:26 [notice] 1#1: start worker process 35
2026/03/19 04:54:26 [notice] 1#1: start worker process 36
2026/03/19 04:54:26 [notice] 1#1: start worker process 37
2026/03/19 04:54:26 [notice] 1#1: start worker process 38
2026/03/19 04:54:26 [notice] 1#1: start worker process 39
2026/03/19 04:54:26 [notice] 1#1: start worker process 40
sristee@Sristee:/mnt/c/Users/srist$ kubectl create deployment web --image=nginx
deployment.apps/web created
```

```
sristee@Sristee:/mnt/c/Users/srist$ kubectl sclae deployment web --replicas=3
error: unknown command "sclae" for "kubectl"

Did you mean this?
        scale
sristee@Sristee:/mnt/c/Users/srist$ kubectl scale deployment web --replicas=3
deployment.apps/web scaled
sristee@Sristee:/mnt/c/Users/srist$ kubectl expose deployment web --port=80 --type=NodePort
service/web exposed
sristee@Sristee:/mnt/c/Users/srist$ kubectl get services
NAME         TYPE        CLUSTER-IP     EXTERNAL-IP   PORT(S)        AGE
kubernetes   ClusterIP   10.43.0.1      <none>        443/TCP        23h
web          NodePort    10.43.48.106   <none>        80:32127/TCP   18s
sristee@Sristee:/mnt/c/Users/srist$ kubectl get pods
NAME                   READY   STATUS    RESTARTS   AGE
nginx                  1/1     Running   0          30m
web-54466ccbf7-49s64   1/1     Running   0          68s
web-54466ccbf7-8f584   1/1     Running   0          68s
web-54466ccbf7-jzpvf   1/1     Running   0          23m
sristee@Sristee:/mnt/c/Users/srist$ kubectl delete pod nginx
pod "nginx" deleted from default namespace
sristee@Sristee:/mnt/c/Users/srist$
```